10.SEASONAL INFLUENZA

Overview

Seasonal influenza, commonly called **flu**, is a contagious respiratory illness caused by influenza viruses. It commonly presents as an acute respiratory illness with **sudden cough, sore throat and fever**, along with symptoms such as body aches, headache and fatigue. Most uncomplicated cases recover with supportive care, but influenza can become severe and lead to complications, particularly in people at higher risk.

Key Facts

- Seasonal influenza is a **contagious respiratory illness** caused by influenza viruses.
- Typical influenza-like illness (ILI) includes **sudden cough and sore throat with measured fever ≥100.4°F**, with onset within the previous 10 days.
- Other symptoms may include:
    - Muscle/body aches
    - Headache
    - Fatigue
    - Runny or stuffy nose
    - Vomiting
    - Diarrhea, particularly in children
- Fever and systemic symptoms usually last around **3 days**, although they may occasionally persist for 5–8 days.
- Cough and fatigue/malaise may persist for **more than two weeks**.
- Complete recovery may take **1–2 weeks or longer**, particularly in older adults.

Symptoms

Common symptoms of influenza include:

- **Fever**
- **Cough**
- **Sore throat**
- Body or muscle aches

- Headache
- Fatigue
- Runny or blocked nose
- Vomiting
- Diarrhea, especially in children

Influenza symptoms usually begin suddenly rather than developing gradually.

Course of Illness

Early illness → Fever + respiratory/systemic symptoms → Recovery

- Fever and systemic symptoms generally last about **3 days**.
- In some patients, symptoms may persist for **5–8 days**.
- Cough and malaise can continue for more than two weeks.
- Older adults may require a longer period for complete recovery.

Patient Categorization

One of the most useful features of the NCDC guideline is its **Category A, B and C** classification.

Category A — Mild / Uncomplicated

Patients may have:

- Mild fever
- Cough and/or sore throat
- With or without:
    - Body aches
    - Headache
    - Diarrhea
    - Vomiting

NCDC guidance:

- Influenza testing is **not required** for Category A patients.
- They generally do **not require oseltamivir**.

- Symptomatic treatment is provided.
- The patient should be monitored and reassessed by a doctor within **24–48 hours**.
- Home isolation and avoidance of contact with the public and high-risk family members are advised.

Category B — More Symptomatic / Higher Risk

Category B includes patients with more significant symptoms and/or those belonging to higher-risk groups.

Important risk factors include:

- High-grade fever
- Severe sore throat
- Persistent cough
- Underlying medical conditions
- Pregnancy
- Young children
- Older adults

Patients in Category B require medical evaluation and appropriate treatment according to clinical assessment.

Category C — Severe Disease

Category C represents patients with **severe symptoms or signs of progressive disease**.

These patients require:

- Immediate medical evaluation
- Appropriate investigations
- Antiviral treatment as clinically indicated
- **Immediate hospitalization** when Category C criteria are met.

Important Warning Signs

Seek urgent medical attention when influenza symptoms are accompanied by:

- Difficulty in breathing or breathlessness
- Severe or progressive illness
- Signs of complications
- Worsening of an underlying chronic disease
- Other clinical features suggesting severe influenza

High-Risk Groups

NCDC identifies the following groups as being at increased risk of complications:

- **Adults ≥65 years**
- Pregnant women, including up to two weeks postpartum
- Children **<5 years**, particularly those <2 years
- People with chronic respiratory disease
- People with chronic heart, kidney, liver or neurological disease
- People with diabetes mellitus
- People with blood disorders, including haemoglobinopathies
- Immunosuppressed individuals, including people living with HIV/AIDS
- People receiving long-term corticosteroids
- Post-transplant patients
- People with extreme obesity (**BMI ≥40 kg/m^2**)

Complications

Severe or progressive influenza can produce a wide range of complications.

In children

Possible complications include:

- Otitis media
- Sinusitis
- Parotitis
- Pneumonia

- Croup
- Bronchiolitis
- Myocarditis
- Rhabdomyolysis
- Encephalopathy/encephalitis
- Invasive bacterial co-infection
- Reye syndrome associated with aspirin use

In adults

Possible complications include:

- Exacerbation of chronic diseases
- Pneumonia
- Acute respiratory distress syndrome (ARDS)
- Bronchitis
- Sinusitis
- Myocarditis
- Pericarditis
- Invasive bacterial co-infection
- Invasive pulmonary aspergillosis

Diagnosis and Testing

Diagnosis begins with **clinical assessment and categorization**.

NCDC's Seasonal Influenza resources include separate guidance for:

- Sample collection
- Laboratory procedures for influenza-virus detection
- Laboratory testing facilities
- Patient reporting
- Influenza surveillance.

Important: Not every patient with influenza-like illness requires laboratory testing. NCDC's patient-categorization guideline specifically states that testing is **not required for Category A patients**.

Laboratory testing is particularly relevant when clinically indicated and for surveillance, severe cases and appropriate public-health investigations.

Treatment and Management

Management depends on the severity and category of illness.

For mild/uncomplicated illness

- Rest
- Adequate fluids
- Symptomatic treatment
- Home isolation
- Monitor symptoms
- Reassessment if symptoms worsen or fail to improve

Category A patients generally do **not require oseltamivir** according to the NCDC categorization guidance.

For higher-risk or severe illness

Patients who fall into higher-risk or severe categories should be assessed by a healthcare professional.

Oseltamivir/antiviral treatment should be used according to NCDC clinical guidance and medical assessment, rather than through self-medication.

Category C patients should be **immediately hospitalized**.

Home Care and Isolation

NCDC's home-care guidance recommends that people with influenza-like illness or appropriate influenza categories:

- Remain **confined at home**.
- Avoid mixing with the public.
- Avoid close contact with high-risk family members.

- Stay in a **well-ventilated room**, preferably separately from other family members.
- Wear a mask as advised.
- Maintain appropriate respiratory hygiene.
- Follow medical advice regarding monitoring and treatment.

NCDC's home-care document advises home isolation for **seven days** and recommends use of a disposable triple-layer mask where available.

Prevention

Prevention focuses on reducing transmission and protecting high-risk individuals.

Personal protection

- Wear a mask when appropriate, particularly around symptomatic individuals.
- Cover the mouth and nose when coughing or sneezing.
- Wash hands regularly.
- Avoid close contact with people who are ill.
- Avoid unnecessary public exposure when experiencing influenza-like symptoms.
- Stay home while recovering according to medical/public-health advice.

Protect high-risk people

People with influenza symptoms should particularly avoid close contact with:

- Older adults
- Infants and young children
- Pregnant women
- People with chronic illnesses
- Immunocompromised individuals

Vaccination

NCDC maintains official guidance on **seasonal influenza vaccine composition and vaccination** and periodically updates recommendations according to the circulating influenza viruses and recommended vaccine composition. The NCDC website currently lists the vaccine composition for the upcoming **2025–2026 season** among its technical resources.

When to Seek Medical Care

Medical evaluation should be sought when:

- Symptoms are severe or worsening.
- There is difficulty breathing or breathlessness.
- Fever or systemic symptoms persist or worsen.
- An underlying chronic disease becomes worse.
- The patient belongs to a high-risk group.
- There are signs of complications.
- The patient is a young child, older adult or pregnant.
- Symptoms do not improve as expected.

Severe/progressive cases should not be managed at home without medical assessment.

Quick Fact Sheet

Feature	**Seasonal Influenza**
Disease	Seasonal influenza (flu)
Cause	Influenza viruses
Main system affected	Respiratory system
Typical onset	Sudden
Major symptoms	Fever, cough, sore throat, body aches, headache, fatigue
Other symptoms	Runny/stuffy nose, vomiting, diarrhea
Typical uncomplicated course	Fever/systemic symptoms ~3 days; recovery generally 1–2 weeks or longer
High-risk groups	Older adults, young children, pregnant women, chronic disease, immunosuppression
Diagnosis	Clinical assessment ± laboratory testing when indicated

Feature	Seasonal Influenza
Treatment	Supportive care; antivirals according to clinical category/medical advice
Prevention	Respiratory hygiene, masking when appropriate, isolation when ill, vaccination
Severe cases	Require urgent medical assessment and may require hospitalization